# ALAN BARNES

(630) 501-6564
Hanover Park, IL
alanbarnes447@gmail.com

www.linkedin.com/in/alanrbarnes
https://github.com/alanrbarnes
https://alanrbarnes.com


## SOFTWARE DEVELOPER

### Superior Computer Programming Skills That Drive Efficiency and Bottom-Line Results

- Execute business systems applications across many business functions; evaluate business and technology requirements, identify best practices, and make recommendations
- Strong foundation in software engineering and programming principles across multiple platforms
- Innovative thinker with ability to envision how technology can be implemented to solve business challenges
- Strong communicator skilled at collaborating successfully with internal business partners and outside vendors
- Excellent troubleshooting skills; able to analyze code and engineer well-researched and cost-effective solutions

- Custom Application Development
- CAD/CAM Software
- Deliver Business Efficiencies
- Requirements Gathering
- Full Stack Programming
- Software Engineering
- Web Design
- Dotnet Programming
- Application Implementation

*"Alan is someone I can give a job to and he will figure out how to do it no matter what." – Owner, Chicago Mold*

## TECHNICAL SKILLS AND AFFILIATIONS

Languages: C#, TypeScript, JavaScript, Python, Java, SQL, HTML, CSS

.NET / Backend: ASP.NET, ASP.NET Core, Entity Framework, RESTful API Design

Frontend: Angular, React, Bootstrap, Tailwind CSS

Databases: SQL Server, MySQL, MongoDB

Cloud & DevOps: Azure Cloud, AWS Cloud, IBM Cloud, Docker, IIS, Git, Postman API Platform

Tools & Environments: Microsoft Visual Studio, Visual Studio Code, Windows, Linux, Mac OS X

Additional Platform Experience: PHP, Laravel, Node.js, Express, Electron, WordPress, Umbraco, Joomla, Ektron, Moodle

Affiliations: Institute for Electrical and Electronics Engineers, Association for Computing Machinery

## PROFESSIONAL EXPERIENCE

Senior Level Programmer/ Web Developer, Living Word Christian Center, Forest Park, IL — 2021 – present
*Globally televised Christian Church, business education leader, and publisher*

Snapshot: Responsible for developing web applications, mobile applications, ecommerce development, server administration, cloud resource administration, database administration, and automation of daily tasks to lower labor costs. Distinct agility in finding robust solutions to diverse use cases and finding useful off the shelf software to save time in development. Optimizing efficiency through setting clear goals, prioritizing tasks, maintaining schedules, breaking tasks into small manageable pieces, and effective time management. Pushing innovation by using new leading-edge technologies and constant continuing education. Self-starter with strong work ethic; administrated numerous server migrations and cloud resource usage changes resulting in tens of thousands of dollars saved.

- Administration of cloud technologies used; Azure, AWS, and IBM cloud, including cloud computing, cloud storage, and cloud networking.
- Migrated applications and Windows services from 3 legacy Rackspace dedicated servers to 2 modernized, actively maintained servers, decommissioning the outdated infrastructure and cutting hosting costs by $3,000/month (~$36,000 annually).

*(Continued: Senior Level Programmer/ Web Developer, Living Word Christian Center)*

- Administration of Cloud Security procedures such as firewalls, IAM, encryption, and SSL
- Developing and publishing apps using Azure cloud services API, Virtual Cloud Servers, and Dedicated Servers
- Administration of database management systems (DBMS) such as MySQL and SQL Server
- Consolidated fragmented hosting infrastructure — merging shared/PHP hosting accounts and standalone virtual machines onto a single dedicated server — reducing infrastructure sprawl and simplifying ongoing maintenance.
- Maintain and support 30+ websites across 100+ domains for a dual-purpose media/publishing and ministry organization, including 3 ASP.NET Core/Angular applications, 5 Ektron (classic ASP.NET) applications, 1 Umbraco site, and numerous WordPress/HTML sites.
- Developed and Published Android and IOS apps using android studio and XCode
- Built and maintained RESTful APIs powering 3 ASP.NET Core/Angular applications, and worked with GraphQL endpoints connecting WordPress-based content to front-end applications.
- Collaborated with cross-functional teams, including end customers, designers and graphic artists to deliver high-quality web solutions
- Implemented responsive design principles to ensure websites are optimized for various devices
- Conducted code reviews and provided constructive feedback to peers to maintain code quality and adherence to coding standards
- Implemented performance optimization techniques to improve website loading times and user experience.

CNC Programmer and Production Supervisor, Chicago Mold, St Charles, IL 2000 – 2021

*Leading manufacturer of plastic injection and compression molds for thermoset and thermoplastic industries*

Mold Maker (2000 – 2003)
CNC Programmer and Production Supervisor (2003 – 2021)

Snapshot: Responsible for analyzing job orders, performing calculations, and simplifying tooling time for workers to increase productivity for clients. Keen ability to use creativity to solve complex problems and to create custom technologies to improve production. Drive productivity of production team through team, leadership, business, manufacturing, and technical knowledge. Demonstrated initiative using C#, plugins, script macros, and JavaScript. Self-starter with strong work ethic; initiated numerous machine improvements resulting in thousands of dollars and hours saved through automation. Directly manage two employees while overseeing a staff of 75.

- Record of successes devising manufacturing and plant operating strategies and continuous improvement projects that automate processes/systems, and deliver productivity, quality and efficiency improvements
- True “jack of all trades”, often seamlessly juggling multiple roles and sets of responsibilities at one time
- Develop department process improvements including eliminating antiquated machines and upgrading to automated machines that do not require operator oversight, work at higher speed, and run overnight; increased production capabilities by 75%
- Technical problem-solver; use varied computer languages including Java Script to automate tedious and time-consuming tasks
- Created application in C# used to interface with the client side of the cam application to include special functionality and automate redundant tasks
- Implemented use of new off-the-shelf part fixturing system that increased the speed at which parts could be setup and machined
- Created post processor used with cam software to automate machines; wrote numerous script-based macros to increase productivity

*(Continued: CNC Programmer and Production Supervisor, Chicago Mold)*

- Designed algorithm-based method for picking up rotational centers on parts by synchronizing computer with the machine; eliminated the need to buy a $20,000 probing system
- Oversee 2 employees; distribute work assignments while providing coaching, feedback, and training
- Completed CAD, CAM, and mechanical design classes to learn how to automate machinery and build tools

Workgroup Member, Institute for Electric and Electronic Engineers, Chicago, IL 2019 – 2021
*Leading developer of industry standards on a broad range of technologies*

Snapshot: Collaborate with stakeholders from IU Health, Medtronic, Draeger Medical Systems, and Columbia University to develop standards for IOT device manufacturers to ensure that new medical devices can connect to the internet securely and to limit unauthorized access.

- Utilize knowledge of database technologies to devise solutions for ensuring confidentiality and functionality of medical device data
- Researched existing privacy and security frameworks used currently in conjunction with the internet and internet related devices to help inform protocols to incorporate greater privacy

## ACADEMIC PROJECTS

Snapshot: Complete numerous development, coding, programming, and web design projects as part of degree coursework. Gathered requirements, translated requirements to code, and collaborated with team members on team-based projects. Full list of projects: https://github.com/alanrbarnes

- Created text analysis software project using C# to read text from a file and scan for complex information; this time saving application enables users to locate information quickly and easily in large bodies of text
- Developed personal project organizer using MongoDB, Express, Node, JavaScript, HMTL and CSS; this universally adaptable software enables users to organize different steps of a project and tasks within each step
- Built personal website as part of web programming and web design classes; utilized HTML, CSS, PHP and Bootstrap
-

## EDUCATION

Bachelor of Science (B.S.), Computer Science, University of Illinois, Springfield, IL December 2020

Cumulative GPA: 3.51/4.00

Coursework: Algorithms & Computations, Data Structures and Algorithms, Software Engineering, Web Development & Programming, Web Design, Applied Statistics, Intro to Operating Systems, Intro to Database Systems, Web Design, Programming Languages, Java Programming

Professional Development & Continuing Education:

- Completed 100+ hours of advanced training in C#, ASP.NET Core, JavaScript, Angular, React, SQL Server, and cloud architecture though technical books, Udemy, and extensive self-directed study
- Engage in ongoing professional development focused on modern software architecture, design patterns, and secure API development.
- Continuously evaluate emerging technologies, frameworks, and development practices to support modern web application development.